RESEARCH ARTICLE

Systemic Inflammation and Recurrence After Atrial Fibrillation–Related Stroke

An Individual Participant Data Meta-Analysis

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Abstract

Background and Objectives

The residual recurrence risk after atrial fibrillation (AF)–related stroke is high despite anticoagulation, thereby necessitating new therapies. The importance of inflammation in AF is increasingly recognized. However, patients with AF-related stroke were excluded from recent trials of anti-inflammatory therapies. It is uncertain whether associations between IL-6/high-sensitivity C-reactive protein (hsCRP) and poststroke recurrence are modified by AF status. In this study, we aimed to analyze the association between IL-6/hsCRP and recurrence according to AF history.

Methods

We leveraged individual participant data from studies identified by systematic review. We analyzed associations between IL-6/hsCRP and recurrent stroke/major adverse cardiovascular events (MACEs) (defined as fatal or nonfatal recurrent stroke or major coronary events) using multivariable Cox regression analyses (conditional logistic regression for 1 study) adjusted for age, sex, index event, cardiovascular risk factors, and medication use, stratified by AF status.

Results

Data from 11 prospective studies with 10,080 patients (2,134 with AF, mean age 70 years, 59.3% male) were included. During 21,080 person-years of follow-up, 1,677 patients had MACEs and 1,342 had recurrent stroke. Inflammatory markers were higher in patients with AF, irrespective of stroke severity and timing of measurement, with elevated levels persisting well beyond the acute phase. hsCRP was associated with recurrent MACEs in patients with AF (adjusted risk ratio [aRR] 1.14, 95% CI 1.04–1.25, per $\log_e$-unit increase) and without AF (aRR 1.08, 1.03–1.13) ($P_{interaction}$ 0.30). There were similar associations on per-quarter analysis in patients with AF (aRR 1.51, 1.04–2.18) and without AF (aRR 1.32, 1.10–1.59) (Q4 vs Q1). No association was observed between hsCRP and recurrent stroke in either group. For IL-6, there was no evidence of interaction according to AF status for MACEs ($P_{interaction}$ 0.57) or recurrent stroke ($P_{interaction}$ 0.82). The aRR per $\log_e$ unit for MACE was 1.17 (1.07–1.27) in patients without AF and 1.10 (0.91–1.34) in those with AF. The corresponding aRR for recurrent stroke was 1.15 (1.05–1.25) and 1.12 (0.91–1.37) in patients without and with AF, respectively.

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Glossary

AF = atrial fibrillation; **AFDAS** = AF detected after stroke; **aRR** = adjusted risk ratio; **CRP** = C-reactive protein; **HR** = hazard ratio; **hsCRP** = high-sensitivity CRP; **IPD** = individual participant data; **IQI** = interquartile interval; **MACE** = major adverse cardiovascular event; **NIHSS** = NIH Stroke Scale; **NLRP3** = NOD-like receptor family pyrin domain-containing 3; **OAC** = oral anticoagulant; **OR** = odds ratio; **PROGRESS** = Perindopril Protection Against Recurrent Stroke Study; **QUIPS** = Quality In Prognosis Studies; **RCT** = randomized controlled trial; **RR** = risk ratio.

Discussion

These data highlight the importance of inflammatory mechanisms in vascular recurrence irrespective of AF, provide rationale for the inclusion of patients with AF in trials of anti-inflammatory therapy, and support a selection approach in future trials based on elevated inflammatory marker levels, rather than stroke etiology alone.

Introduction

The global burden of stroke continues to increase despite the availability of effective secondary prevention therapies,[1] underscoring the urgency of identifying new targets for stroke prevention. Atherosclerosis is a chronic systemic inflammatory disorder, and anti-inflammatory therapies have emerged as a therapeutic strategy for cardiovascular disease.[2] Colchicine lowers the risk of major adverse cardiovascular events (MACEs) in patients with coronary artery disease.[3] Canakinumab has also demonstrated efficacy in reducing residual vascular risk through inhibition of interleukin-1β (IL-1β) in patients with elevated C-reactive protein (CRP).[4] Inflammatory signaling is clearly and unequivocally implicated in the pathogenesis of stroke due to atherosclerotic mechanisms as evidenced by supporting data from clinicopathologic studies,[5] animal models,[6,7] genetic epidemiology,[8,9] and large observational studies.[10-13] These data underpinned the rationale for the design of the first randomized controlled trials (RCTs) of anti-inflammatory therapies for secondary prevention after stroke, both of which excluded patients with cardioembolic stroke.[14,15] However, future RCTs are now being planned, and it remains uncertain whether restriction of trial eligibility to patients with non-cardioembolic or atherosclerotic stroke subtypes is justified.

Patients who experience a stroke due to atrial fibrillation (AF) remain at a considerable risk of future events, even with oral anticoagulation.[16] This suggests that current prevention strategies are inadequate and new therapeutic strategies beyond anticoagulation are warranted. Multiple lines of evidence have recently indicated that inflammation plays a key role in the pathogenesis of AF and cardioembolism. Experimental data indicate that the NOD-like receptor family pyrin domain-containing 3 (NLRP3) inflammasome is upregulated in cardiomyocytes in AF and promotes structural and electrical atrial remodeling.[17] Inflammation is associated with a lifetime risk of incident AF,[9,18,19] AF-related stroke,[20] and cardioembolic stroke.[19] However, there are limited data on the association between inflammation and prognosis after ischemic stroke in patients with AF. Therefore, in this individual participant data (IPD) meta-analysis, we investigated the association between high-sensitivity CRP (hsCRP), interleukin-6 (IL-6), and vascular recurrence after stroke according to AF status.

Methods

Systematic Review

We conducted a systematic review to identify studies investigating the association between inflammatory markers and poststroke vascular recurrence. The results of the systematic review were previously published,[21] and the search was since updated.[13] Eligible studies were as follows: (1) included patients with a history of ischemic stroke or TIA; (2) measured IL-6 or hsCRP after stroke; (3) provided outcome data for recurrent stroke or MACEs during follow-up; (4) were either observational cohorts, case-cohort studies, or observational data from RCTs. Unpublished studies were also identified through correspondence with experts. The study protocol was published in PROSPERO (ID CRD42018116190).

Individual Participant Data Meta-Analysis

Authors of studies identified by systematic review were then invited to participate in an IPD meta-analysis—the Blood Inflammatory markers in Stroke Collaboration (BISC)—and the main findings have been published.[22] Participating studies sent IPD to the coordinating center in Dublin, Ireland. Data were harmonized using standardized definitions of covariates and outcome events (eMethods). This article complies with Preferred Reporting Items for Systematic Reviews and Meta-Analyses of Individual Participant Data recommendations.[23]

AF

For this analysis, patients were grouped into those with and without AF at baseline, defined as a history of, or a new diagnosis of, paroxysmal, persistent, or permanent AF during the treatment episode for the index stroke. Data were not available to differentiate between known AF before stroke and AF detected after stroke (AFDAS).

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Outcome Events

The co-primary end points in BISC were prespecified and included the following: (1) any MACE and (2) recurrent stroke during follow-up. MACE was predefined as any recurrent stroke, major coronary event, or death due to a vascular cause. Recurrent stroke included any ischemic, hemorrhagic, or unknown stroke type (eTable 1). Only outcome events that occurred after inflammatory marker measurement were included.

Study Quality

The Quality In Prognosis Studies (QUIPS) tool was used in the evaluation of study quality (eMethods).[24]

Statistical Analysis

Between-group differences in inflammatory marker levels according to stroke severity were assessed using the Wilcoxon rank-sum test, with nonsevere stroke defined as a NIH Stroke Scale (NIHSS) score of 4 or less. The relationship between levels of poststroke IL-6 and hsCRP and the timing of measurement in patients with and without AF was analyzed by plotting their median (interquartile interval [IQI]) concentrations within each quarter of the sampling time frame. Between-group differences in biomarker concentrations according to timing of measurement were then analyzed at each time point using the Wilcoxon rank-sum test. We then explored the association between each inflammatory marker and recurrent vascular events. For studies with time-to-event data, we used Cox proportional hazards regression to estimate the hazard ratio (HR) for the outcomes of recurrent stroke and MACE. For the 1 nested case-control study (Perindopril Protection Against Recurrent Stroke Study [PROGRESS]), we performed conditional logistic regression to estimate the odds ratio (OR) for both outcomes. IL-6 and hsCRP were the exposure variables of interest for all analyses. Both HRs and ORs were assumed to approximate the relative risk (because the outcomes are rare) and were collectively summarized as risk ratios (RRs) with 95% CIs. Inflammatory marker distributions were positively skewed and underwent logarithmic transformation. The proportional hazards assumption was analyzed by Schoenfeld tests. Associations with recurrent vascular events were analyzed per quarter and per-unit increase in $\log_e$-transformed IL-6 and hsCRP. A single-unit increase in IL-6 and/hsCRP after $\log_e$ transformation approaches a 3-fold increase on a natural scale. We performed unadjusted analyses and analyses adjusted for age, sex, index event, hypertension, smoking, diabetes mellitus, coronary artery disease, statin therapy, antithrombotic use, and trial arm (in the single study [PROGRESS] that provided observational data from an RCT). Ten studies provided directly available IPD. One study (PROGRESS) analyzed IPD locally using identical statistical code because of restrictions on data transfer, which precluded centralized analysis; these results were pooled with the other 10 studies using random-effects meta-analysis with inverse variance weighting. Subgroup analyses were performed according to the AF status. Effect modification of hsCRP and IL-6 by AF status was assessed using a test for interaction. Only outcome events that occurred after sample measurement were considered. We then performed an analysis for the association with recurrent ischemic stroke, which was not prespecified. Finally, we performed sensitivity analyses for the primary outcome after removal of the single study without time-to-event data and additional analyses that adjusted for anticoagulant use. Analyses were performed using Stata Release 17.

Standard Protocol Approvals, Registrations, and Patient Consents

Because this was an individual participant data meta-analysis of completed cohort studies, informed consent to participate was not required. Each included study obtained ethical approval from their local institutional review board.

Data Availability

Data requests will be considered after agreement with the Steering Committee, provided data transfer complies with general data protection regulation and is approved by the respective Ethical Review Board.

Results

Study Selection

Using the search strategy, we identified 3,885 records, and 34 studies met the eligibility criteria. Following invitations to authors of each study, investigators from 11 studies agreed to participate in a network meta-analysis (PRISMA flowchart presented in eFigure 1). There were 10 prospective cohort studies and 1 nested case-control study within an RCT (PROGRESS) (details of each study are summarized in eTable 2). Overall, 10,080 patients were included.

Study Quality

No study had a high risk of bias in any of the 6 domains of the QUIPS tool (eTable 3). A detailed description of each study and the rationale for quality assessment scoring are provided in eMethods.

Clinical Characteristics

The baseline characteristics of participants are summarized in Table 1. Data for each individual study are also presented in eTable 4. Of the 10,080 patients, 8,574 (85%) had experienced an ischemic stroke, 5,974 (59.3%) were male, and 2,134 (21.2%) had AF. Data were available for hsCRP in 9,797 patients and IL-6 in 6,753 patients. The median time from the qualifying event to inflammatory marker measurement was 1 day (IQI 0.5–4) in the 10 studies in which IPD were directly available. In PROGRESS, samples were measured in the convalescent phase, with a median time to phlebotomy of 310 days (IQI 116–760). Compared with patients without AF, patients with AF were older, were more frequently female, and had more severe stroke (eTable 5). Oral anticoagulant (OAC) use at discharge in patients with AF varied considerably across studies (median proportion use 68.9%, range 28.9%–94.7%), which likely reflected lower use in older

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Table 1 Baseline Characteristics

	Studies with directly available IPD (n = 8,502)	PROGRESS (n = 1,578)
Age	72 (62–81)	67 (61–73)
Male sex	4,807 (56.5)	1,167 (74.0)
Index event TIA	1,174 (13.8)	332 (21.0)
NIHSS scores	4 (1–12)	N/A
Hypertension	6,172 (72.6)	934 (59.2)
Atrial fibrillation	1996 (23.5)	138 (8.7)
Diabetes mellitus	1,598(18.9)	200 (12.7)
Current smoker	1906 (22.5)	314 (19.9)
Coronary artery disease	1,400 (16.6)	292 (18.5)
Hyperlipidemia	5,520 (65.9)	1,366 (86.6)
Antithrombotic therapy	6,245 (72.8)	1,393 (88.3)
Oral anticoagulants at discharge	1,325 (16.7)	
Statins	4,571 (56.0)	123 (7.8)
IL-6 (pg/mL)	4.5 (2.11–12.9)	2.3 (1.5–3.5)
hsCRP (mg/L)	2.88 (1.09–7.78)	2.0 (0.9–4.6)
Time to sample measurement (d)	1 (0.5–4)	310 (116–760)

Abbreviations: BISC = Blood Inflammatory markers in Stroke Collaboration; IPD = individual participant data; N/A = not available; NIHSS = NIH Stroke Scale; PROGRESS = Perindopril Protection Against Recurrent Stroke Study; TIA = transient ischemic attack.
Categorical variables are presented as number (%) and continuous variables as median (IQI).

studies. After exclusion of patients with missing data, 909 patients with AF (66.0%) were prescribed OACs at discharge. In the AF subgroup, the index stroke mechanism was large artery atherosclerosis in 90 (4.2%), small vessel occlusion in 56 (2.6%), cardioembolism in 1,526 (72.0%), undetermined in 441 (20.8%), and other determined etiology in 7 (0.3%). In patients without AF, the index mechanism was due to large artery atherosclerosis in 1,626 (20.6%), small vessel occlusion in 1,304 (16.5%), cardioembolism in 1,105 (14%), undetermined in 3,626 (45.9%), and other determined cause in 235 (3.0%) (eTable 6).

Peripheral Inflammation According to AF Status and Stroke Severity

IL-6 levels were substantially higher in patients with AF (7.46 pg/mL, IQI 3.46–17.5) than in those without AF (3.98 pg/mL, IQI 1.92–11, $p < 0.001$) (Figure 1A). Similar differences were observed for hsCRP (4.17 mg/L, IQI 1.76–11.84, vs 3.0 mg/L, IQI 1.11–6.53, $p < 0.001$) (Figure 1B). Higher levels of both markers were also observed in patients with AF in analyses restricted to nonsevere stroke or TIA. IL-6 levels were 5.69 pg/mL (IQI 2.48–11.23) in patients with AF and 3.08 pg/mL (IQI 1.5–7.17) in those without AF ($p < 0.001$) when patients with an NIHSS score ≥5 were excluded (Figure 1C). The corresponding levels for hsCRP were 3.46 mg/L (IQI 1.4–9.51) and 2.29 mg/L (IQI 0.96–5.8) ($p < 0.001$) (Figure 1D). Again, when the analysis was restricted to patients with TIA, a similar pattern was observed, both for IL-6 (4.24 pg/mL, IQI 1.69–8.04, vs 2.65 pg/mL, IQI 1.47–6.22, $p = 0.003$) and hsCRP (3.01 mg/L, IQI 1.15–8.93, vs 2.27, IQI 0.95–6.08, $p = 0.02$) (Figure 1, E and F).

Peripheral Inflammation According to AF Status and Timing of Measurement

We observed higher levels of hsCRP in patients with AF compared with patients without AF at all time points from the hyperacute to postacute phases (Figure 2A). For IL-6, levels were similar in both groups when measured in the first 24 hours after the index event but were significantly higher in patients with AF thereafter at all other time points, including the postacute phase (Figure 2B). In the single study (PROGRESS) that measured inflammatory markers exclusively in the convalescent phase, both IL-6 (2.78 pg/mL [IQI 1.77–4.00] vs 2.24 pg/mL [IQI 1.48–3.44], $p = 0.003$) and hsCRP (2.71 mg/L [IQI 1.26–5.44] vs 1.92 mg/L [IQI 0.88–4.53], $p = 0.01$) remained substantially higher in the AF subgroup at an average of almost 1 year after stroke.

Follow-Up

The total cumulative duration of follow-up was 21,085 person-years. During this period, 1,677 patients (16.6%, 95% CI 15.9–17.4) experienced at least 1 MACE. Of these, the first event was a recurrent stroke in 1,338, a major coronary event in 311, or

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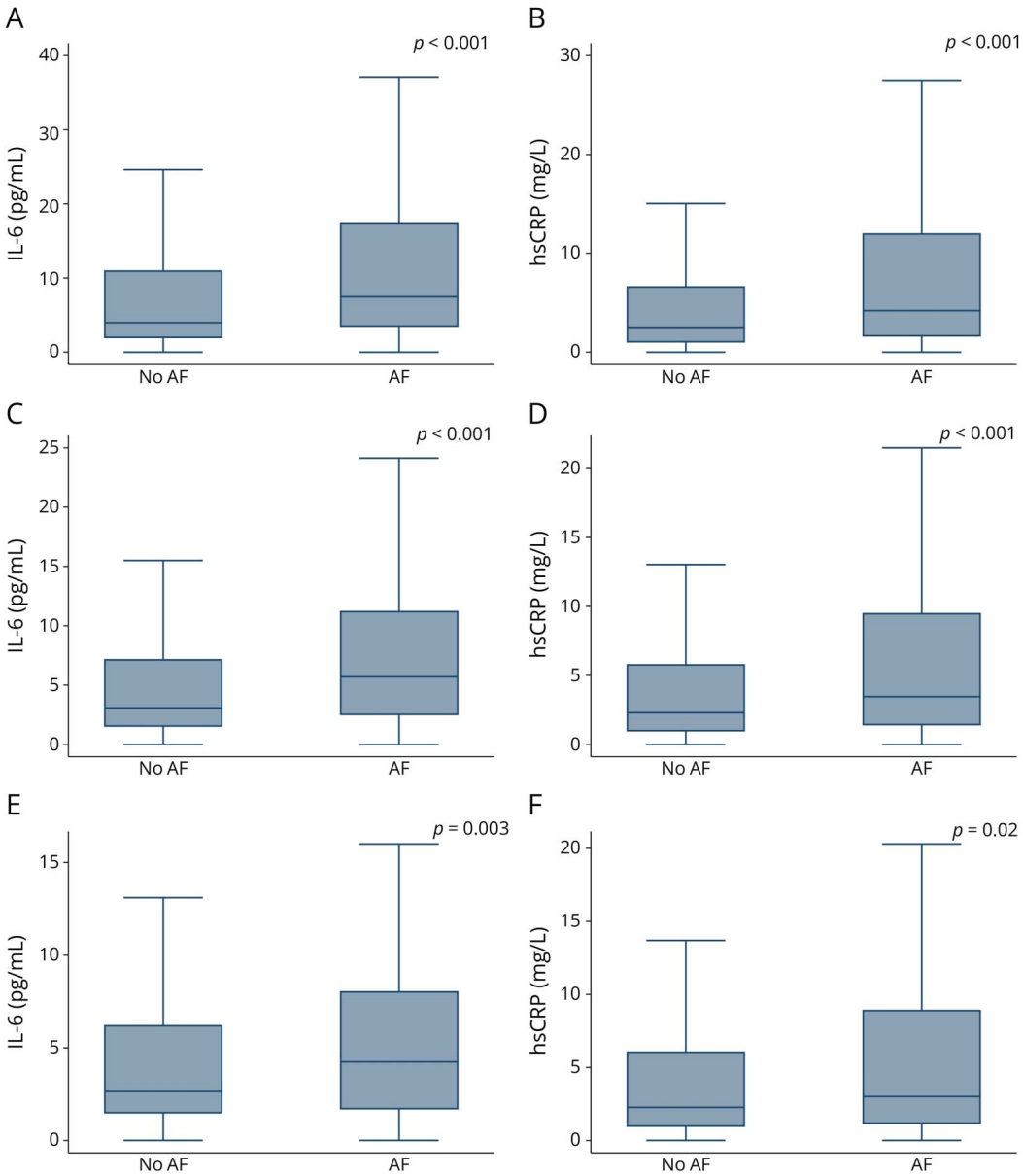
Figure 1 Inflammatory Marker Levels According to Stroke Severity and AF Status



Data are shown for (A and B) IL-6 and hsCRP in all ischemic stroke or TIA events; (C and D) IL-6 and hsCRP restricted to nonsevere stroke (NIHSS score ≤4) or TIA; and (E and F) IL-6 and hsCRP in TIA only. AF = atrial fibrillation; hsCRP = high-sensitivity CRP; NIHSS = NIH Stroke Scale.

vascular death of unspecified cause in 28. In total, 1,342 participants (13.3%, 95% CI 12.7–14.0) experienced a recurrent stroke during follow-up (1,220 ischemic, 111 hemorrhagic, and 11 unknown pathologic subtype).

hsCRP and Recurrent Vascular Events According to AF Status

In unadjusted analysis, increasing circulating levels of hsCRP were associated with MACEs in patients with AF (RR 1.17, 95% CI 1.08–1.27, per $\log_e$ unit) (Table 2). In per-quarter analysis, the RR for patients in the top vs bottom 25% of hsCRP levels was 1.72 (95% CI 1.21–2.46; Figure 3). After adjustment for age, sex, cardiovascular risk factors, and secondary prevention medication use, the association was attenuated but remained statistically significant (RR 1.14, 95% CI 1.04–1.25, per $\log_e$ unit). The risk of MACE was approximately 50% higher in patients with AF who had hsCRP levels in the top vs bottom quarter (adjusted RR 1.51, 95% CI 1.04–2.18; Figure 3). In patients without AF, the association with MACE was similar to that in patients with AF, without evidence of interaction ($P_{interaction} = 0.30$). The adjusted RR was 1.08 (95% CI 1.03–1.13) per $\log_e$-unit increase, and the corresponding RR was 1.32 (95% CI 1.10–1.59, Q4 vs Q1) (Figure 3). Associations with recurrence increased in a dose-dependent manner irrespective of AF status.

For the outcome of recurrent stroke, no association was observed for hsCRP in patients with or without AF (Table 3). The adjusted RR for recurrent stroke in patients with AF was

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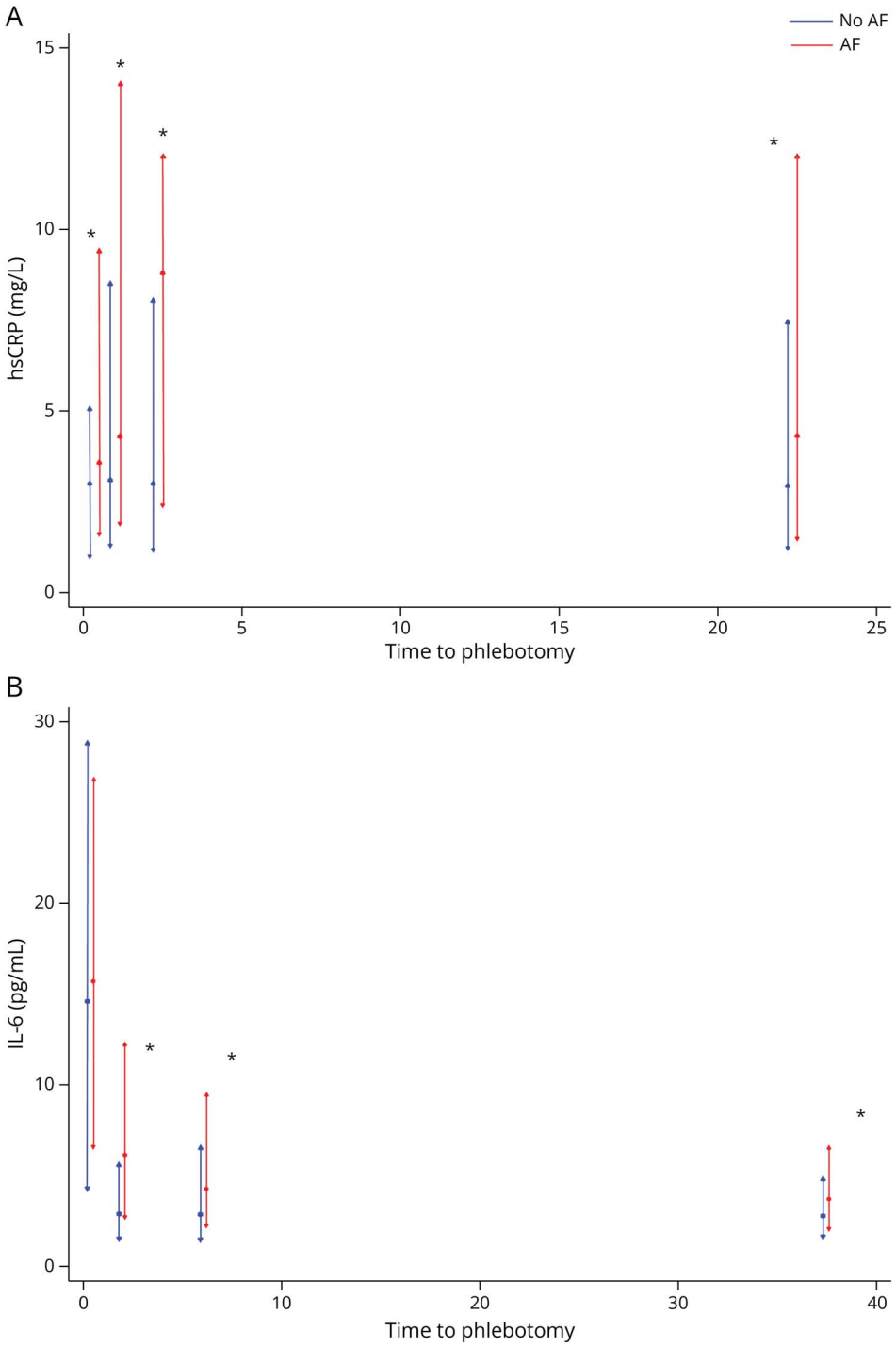
Figure 2 Poststroke Dynamics of (A) hsCRP and (B) IL-6 Concentrations



The median (IQI) concentrations of each marker were plotted within each quarter of the sampling time frame. Between-group differences at each time point were analyzed by the Wilcoxon rank-sum test. *Statistically significant difference (p 0.05) in biomarker concentrations. hsCRP = high-sensitivity CRP; IQI = interquartile interval.

1.05 (0.94–1.17) per $\log_e$ unit and was 0.99 (95% CI 0.94–1.05) in patients without AF (Figure 3). There was no evidence of an interaction between AF and hsCRP ($p = 0.35$).

IL-6 and Recurrent Vascular Events According to AF Status

In unadjusted analysis, IL-6 was associated with MACEs during follow-up in patients without AF, both per $\log_e$ unit (RR 1.25, 95% CI 1.16–1.35) and per quarter (RR 1.98, 95% CI 1.58–2.48, Q4 vs Q1) (Table 2, Figure 3). The corresponding RRs in patients with AF were similar but did not reach statistical significance (RR 1.15, 95% CI 0.98–1.36 per $\log_e$ unit and RR 1.78, 95% CI 0.93–3.39 [Q4 vs Q1]). Similar findings were observed after adjustment for cardiovascular risk factors and medication use. However, there was no evidence of an interaction between AF status and MACE for IL-6

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Table 2 IL-6, hsCRP, and Recurrent MACEs in Patients With and Without Atrial Fibrillation

Marker	Model	Exposure	Subgroup	RR (95% CI)	*p* Value	N (n)	Subgroup	RR (95% CI)	*p* Value	N (n)
IL-6	Unadjusted	Per log unit	No atrial fibrillation	1.25 (1.16–1.35)	<0.001	5,556 (917)	Atrial fibrillation	1.15 (0.98–1.36)	0.10	1,018 (117)
		Q1		Reference				Reference		
		Q2		1.13 (0.92–1.40)	0.24			1.15 (0.59–2.25)	0.68	
		Q3		1.39 (1.12–1.72)	0.003			1.73 (0.95–3.16)	0.07	
		Q4		1.98 (1.58–2.48)	<0.001			1.78 (0.93–3.39)	0.08	
	Adjusted	Per log unit	No atrial fibrillation	1.165 (1.073–1.265)	<0.001	5,216 (858)	Atrial fibrillation	1.10 (0.91–1.34)	0.31	906 (92)
		Q1		Reference				Reference		
		Q2		1.04 (0.83–1.30)	0.71			0.88 (0.42–1.83)	0.73	
		Q3		1.17 (0.93–1.47)	0.18			1.58 (0.83–3.01)	0.17	
		Q4		1.62 (1.27–2.07)	<0.001			1.32 (0.63–2.74)	0.47	
hsCRP	Unadjusted	Per log unit	No atrial fibrillation	1.09 (1.05–1.15)	<0.001	7,658 (1,275)	Atrial fibrillation	1.17 (1.08–1.27)	<0.001	1928 (294)
		Q1		Reference				Reference		
		Q2		0.99 (0.83–1.18)	0.92			0.99 (0.65–1.50)	0.97	
		Q3		1.11 (0.94–1.33)	0.23			1.34 (0.91–1.96)	0.14	
		Q4		1.40 (1.18–1.67)	<0.001			1.72 (1.21–2.46)	0.003	
	Adjusted	Per log unit	No atrial fibrillation	1.08 (1.03–1.13)	0.001	7,298 (1,211)	Atrial fibrillation	1.14 (1.04–1.25)	0.005	1808 (264)
		Q1		Reference				Reference		
		Q2		1.03 (0.86–1.24)	0.75			0.91 (0.59–1.39)	0.65	
		Q3		1.10 (0.92–1.32)	0.30			1.19 (0.81–1.77)	0.38	
		Q4		1.32 (1.10–1.59)	0.003			1.51 (1.04–2.18	0.03	

Abbreviations: AF = atrial fibrillation; N = total cases; n, events; Q = quarter; RR = risk ratio.
Analyses adjusted for age, sex, index event, hypertension, smoking, diabetes mellitus, coronary artery disease, statin therapy, antithrombotic use, and trial arm (where indicated).

($P_{interaction}$ = 0.57) on formal statistical testing. In the AF subgroup, the adjusted RR was 1.32 (95% CI 0.63–2.74, Q4 vs Q1), and in patients without AF, the RR was 1.62 (95% CI 1.27–2.07, Q4 vs Q1). Again, associations between inflammatory markers and MACEs seemed to increase in a dose-dependent manner.

A statistically significant association with recurrent stroke was seen for IL-6 in patients without AF (adjusted RR 1.15, 95% CI 1.05–1.25, per $\log_e$-unit analysis and RR 1.56, 95% CI 1.21–2.02 [Q4 vs Q1]), but not in patients with AF (RR 1.12, 95% CI 0.91–1.37 per unit, RR 1.23, 95% CI 0.57–2.69, Q4 vs Q1) (Figure 1). However, again, no evidence of an interaction was observed according to the AF subgroup (p = 0.82).

Secondary Outcomes

We performed an additional analysis for the outcome of recurrent ischemic stroke (eTable 7). IL-6 was associated with recurrent ischemic stroke in patients without AF (adjusted risk ratio [aRR] 1.17, 1.06–1.28) and showed a directionally

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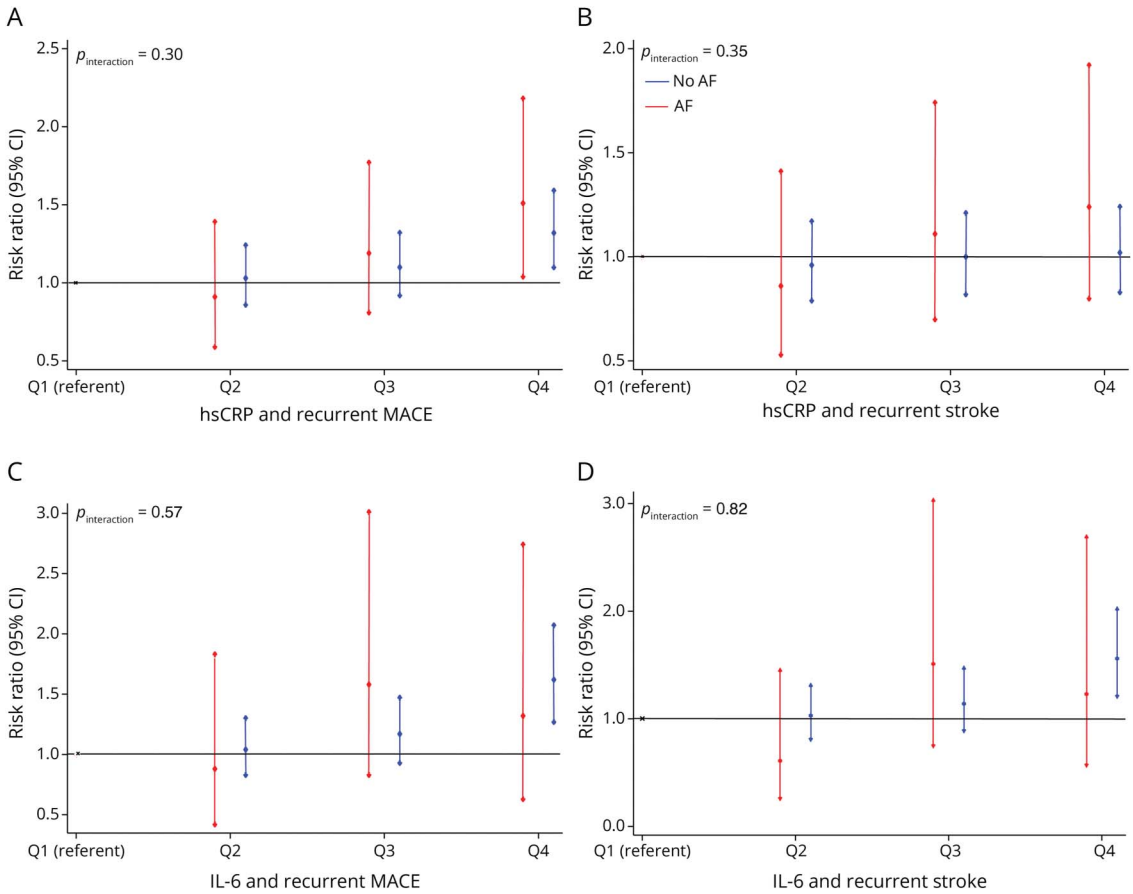
Figure 3 Inflammatory Markers and Risk of Outcome Events According to AF Status



(A) hsCRP and MACE; (B) hsCRP and recurrent stroke; (C) IL-6 and MACEs; (d) IL-6 and recurrent stroke. Analyses are adjusted for age, sex, index event, cardiovascular risk factors, and secondary prevention medication use. hsCRP = high-sensitivity CRP; MACE = major adverse cardiovascular event.

similar nonsignificant association in patients with AF (aRR 1.13, 0.92–1.39). However, there was no evidence of a between-group interaction (p = 0.76) to suggest that associations differed by AF status. No association with recurrent ischemic stroke was observed for hsCRP irrespective of AF.

Sensitivity Analyses

We performed a sensitivity analysis to explore the influence of the single study without time-to-event data (PROGRESS) on our results. However, removing the study from the analysis did not influence our findings (eTables 8 and 9). We observed independent associations with MACE, but not recurrent stroke, for increasing hsCRP levels irrespective of AF status. We also found that IL-6 was consistently associated with recurrent stroke in patients without AF. Again, no evidence of an interaction was observed for the exposure-outcome relationship according to the AF subgroup for IL-6 and recurrent stroke, although the effect estimate was comparatively weaker and not significant in patients with AF. We performed a sensitivity analysis that adjusted for OAC use. After exclusion of patients without missing data, we observed a consistent association between hsCRP and MACEs in patients with AF (aRR 1.16, 95% CI 1.05–1.28, per $\log_e$ unit). A nonsignificant trend toward association was observed for IL-6 with both MACE (aRR 1.29, 95% CI 0.97–1.72) and recurrent stroke (aRR 1.38, 95% CI 0.98–1.94) (eTable 10).

Discussion

In this large meta-analysis of individual participant data from 11 prospective cohort studies, we have provided important new data on the role of IL-6–CRP signaling in prognosis after AF-related stroke. We have shown that peripheral inflammation is substantially higher in patients with AF compared with patients without AF after stroke, and that this finding occurs irrespective of stroke severity and persists well beyond the acute stroke period. We have shown that inflammation is associated with a higher vascular recurrence risk irrespective of AF status, with the risk increasing in a dose-dependent manner. We observed that hsCRP is independently associated with MACEs during follow-up in patients with AF even after extensive adjustment for cardiovascular risk factors and secondary prevention medication use. Although we did not observe an independent association between IL-6 and vascular recurrence in AF, there was no

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Table 3 IL-6, hsCRP, and Recurrent Stroke in Patients With and Without Atrial Fibrillation

Marker	Model	Exposure	Subgroup	RR (95% CI)	*p* Value	N (n)	Subgroup	RR (95% CI)	*p* Value	N (n)
IL-6	Unadjusted	Per log unit	No atrial fibrillation	1.22 (1.12–1.32)	<0.001	5,556 (763)	Atrial fibrillation	1.16 (0.96–1.39)	0.12	1,018 (93)
		Q1		Reference				Reference		
		Q2		1.09 (0.86–1.37)	0.49			1.01 (0.47–2.18)	0.98	
		Q3		1.30 (1.03–1.64)	0.03			1.72 (0.88–3.38)	0.11	
		Q4		1.84 (1.45–2.33)	<0.001			1.68 (0.82–3.45)	0.16	
	Adjusted	Per log unit	No atrial fibrillation	1.15 (1.05–1.25)	0.002	5,216 (723)	Atrial fibrillation	1.12 (0.91–1.37)	0.29	906 (76)
		Q1		Reference				Reference		
		Q2		1.03 (0.81–1.31)	0.83			0.61 (0.26–1.45)	0.26	
		Q3		1.14 (0.89–1.47)	0.29			1.51 (0.75–3.03)	0.25	
		Q4		1.56 (1.21–2.02)	0.001			1.23 (0.57–2.69)	0.60	
hsCRP	Unadjusted	Per log unit	No atrial fibrillation	0.99 (0.95–1.04)	0.78	7,658 (1,039)	Atrial fibrillation	1.08 (0.98–1.20)	0.13	1928 (198)
		Q1		Reference				Reference		
		Q2		0.91 (0.76–1.10)	0.35			0.92 (0.57–1.49)	0.74	
		Q3		0.99 (0.82–1.19)	0.87			1.19 (0.77–1.86)	0.44	
		Q4		1.05 (0.87–1.27)	0.63			1.39 (0.91–2.10)	0.13	
	Adjusted	Per log unit	No atrial fibrillation	0.99 (0.94–1.05)	0.84	7,298 (997)	Atrial fibrillation	1.05 (0.94–1.17)	0.39	1808 (179)
		Q1		Reference				Reference		
		Q2		0.96 (0.79–1.17)	0.69			0.86 (0.53–1.41)		
		Q3		1.00 (0.82–1.21)	0.99			1.11 (0.70–1.74)		
		Q4		1.02 (0.83–1.24)	0.88			1.24 (0.80–1.92)		

Abbreviations: n = events; N = total cases; Q = quarter; RR = risk ratio.
Analyses adjusted for age, sex, index event, hypertension, smoking, diabetes mellitus, coronary artery disease, statin therapy, antithrombotic use, and trial arm (where indicated).

evidence to support a clinically significant interaction according to AF status.

Inflammation is a new target for cardiovascular prevention. However, perhaps owing to a large body of experimental evidence indicating that immune mechanisms are central to plaque progression, most of the translational potential of anti-inflammatory therapies has focused on atherosclerosis. The Canakinumab Anti-inflammatory Thrombosis Outcomes Study (CANTOS) trial provided proof of concept that inhibition of the IL-1β-IL-6-CRP axis with canakinumab reduced the risk of MACEs in coronary atherosclerosis.[4] These data were backed up by observational[12,25,26] and genetic epidemiologic studies,[9,27] showing that IL-6–CRP signaling is implicated in incident and recurrent coronary events. In stroke, elevated IL-6 and hsCRP are associated with a higher risk of incident stroke and recurrent vascular events, and these associations are independent of established cardiovascular risk factors.[10,12,22] Mendelian randomization studies have also shown that IL-6 signaling is likely to be causally mediated in

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the lifetime risk of stroke, with subtype-specific associations appearing to be largely limited to large artery atherosclerotic stroke.[9] These data have largely guided trial design to date. Both CONVINCE and CHANCE-3 trials investigated the efficacy of colchicine for secondary prevention after ischemic stroke and excluded cardioembolic stroke.[14,15] These first anti-inflammatory trials for secondary stroke prevention did not demonstrate benefit, which may be due to a number of reasons, including insufficient statistical power or patient selection. Efforts to refine patient selection in future trials of anti-inflammatory therapies will be vital to their success. It is debatable whether eligibility criteria should center on stroke mechanistic subtype or an inflammatory marker–driven selection approach.

The residual risk of recurrent stroke despite anticoagulation after AF-related stroke is high.[16] Current understanding of the factors responsible for this residual risk is insufficient, and new insights into the pathophysiology of AF-related stroke are necessary to tailor appropriate secondary prevention strategies.[28] Several complementary lines of evidence indicate that inflammation promotes and sustains AF.[29] Clinicopathologic studies demonstrate that patients with AF exhibit enhanced NLRP3 inflammasome expression within atrial cardiomyocytes,[30] and experimental data indicate that NLRP3 inflammasome activity promotes arrhythmia inductility and cardiac remodeling.[17] Population-based studies have shown that hsCRP is associated with a higher risk of incident AF[18] and elevated IL-6 levels are independently associated with incident stroke in patients with established AF even after adjustment for cardiovascular risk factors.[20] Genetic epidemiologic data also point to the causal importance of NLRP3-mediated downstream signaling in AF and AF-related stroke, showing that IL-6 lowering is associated with a reduced risk of AF[9] and interleukin-18 (IL-18) activity is associated with cardiac remodeling, AF, and cardioembolic stroke.[19] Taken together with our data, there is robust evidence indicating that activation of the central signaling cascade through NLRP3 and IL-6–hsCRP plays an important role in AF progression and AF-related stroke, may in part explain some of the untreated residual risk in these patients, and could represent a valid treatment target for secondary prevention. It is important to note that these data also make the case for inclusion of patients with AF in future trials of anti-inflammatory therapies for stroke prevention. We have recently reported that associations between circulating IL-6, hsCRP, and vascular recurrence after stroke are not modified by the presence of atherosclerosis, suggesting that systemic inflammation after stroke may be associated with an adverse prognosis irrespective of the underlying stroke mechanism.[31] Recent experimental data have eloquently illustrated the complex bidirectional association between stroke-induced brain injury and chronic alterations in resident innate immune cells across multiple organs, including the heart. IL-1β release after brain infarction causes epigenetic reprogramming in myeloid cells, ultimately leading to cardiac fibrosis, chronic diastolic dysfunction, and AF inducibility, which was reversible on inhibition of IL-1β or blockade of proinflammatory monocyte trafficking.[32] These data would again support that the use of anti-inflammatory therapies after stroke may extend beyond atheroprotective effects.

Our study has strengths and limitations. We cannot be certain that associations between inflammatory markers and recurrence are causal. We performed extensive adjustment for established cardiovascular risk factors, but we did not have data for a number of risk factors known to be associated with AF-related stroke recurrence, including chronic kidney disease, AF burden, congestive cardiac failure, left atrial size, and whether the AF was detected after stroke or known before stroke. We acknowledge that this raises the potential for residual confounding. Although we did not observe evidence of interaction between IL-6 and AF status for the outcome of recurrent MACEs, it does not exclude the possibility that true associations are weaker in patients with AF than in non-AF patients because statistical power may have influenced the sensitivity of the test for interaction. Unfortunately, our results cannot provide sufficient insights to draw firm conclusions regarding the pathophysiologic mechanisms by which inflammation mediates an elevated recurrence risk in AF. However, potential pathways include inflammation-driven atrial thrombosis, cardiac remodeling, AF progression, or recurrence due to atherosclerotic mechanisms. A limitation of this work is that we pooled data from studies with independent protocols, which used a variety of different assays. It would be preferable to replicate our findings using a single standardized commercially available assay. We did not have information on whether AF was known before stroke or detected after stroke (AFDAS), which would have been informative, because AFDAS is now recognized as a distinct clinical entity. We had incomplete data for OAC administration after discharge, and the overall proportion of patients prescribed OAC therapy was lower than would be expected in the current era. However, between-study differences are largely explained by the study period, and the results were unchanged in a sensitivity analysis adjusting for OAC use, suggesting that the role of inflammation in recurrence is independent of OAC therapy. The strengths of this study include a large sample size, access to IPD identified by a comprehensive systematic review of the literature, and harmonization of data across studies to ensure methodological standardization of covariates. Availability of information on timing of sample measurement allowed for insights into the dynamics of IL-6 and hsCRP after stroke in patients with AF. Finally, despite growing evidence indicating the importance of inflammation in AF pathogenesis, very few studies have previously investigated its role after AF-related stroke, and our findings illustrate the importance of IL-6–hsCRP signaling in these patients.

We have provided new evidence demonstrating elevated systemic inflammation after AF-related stroke and data suggesting that known associations between IL-6–hsCRP signaling and recurrence are not modified by the presence of AF.

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These data do not lend support to the exclusion of patients with AF from RCTs of anti-inflammatory therapies after stroke. This analysis also provides rationale for future work investigating the pathophysiologic role of immune mechanisms in vascular recurrence after AF-related stroke.

Author Contributions

J.J. McCabe: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. K. Harris: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. S. Gorey: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. C. Walsh: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. M. Arnold: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. G.M. De Marchis: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. P. Hervella: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. R. Iglesias-Rey: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. C. Jern: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. M. Katan: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. L. Li: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. N. Miyamoto: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. J. Montaner: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. F. Purroy: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. P.M. Rothwell: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. T.M. Stanne: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. C.L.M. Sudlow: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. Y. Ueno: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. M. Vicente-Pascual: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. W.N. Whiteley: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. M. Woodward: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. P.J. Kelly: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data.

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